Introduction

Plagiarism is a major challenge in academic writing and research. Traditional plagiarism detection methods mainly focus on identifying identical or similar words. However, modern plagiarism can involve paraphrasing, sentence restructuring and replacing words with synonyms while retaining the original meaning. This makes plagiarism detection more challenging.

The project **"Explainable Semantic Plagiarism Detection with Internet Source Checking"** aims to develop a system that identifies exact, partial and paraphrased plagiarism using lexical and semantic similarity techniques. The proposed system will use TF-IDF and Sentence-BERT (SBERT) to compare textual passages and identify potentially matching content.

The research is based on the **PAN 2025 plagiarism detection overview paper**, which provides insights into recent plagiarism detection approaches and their performance. For preliminary experiments, the project uses the **PAN 2013 dataset** to compare different similarity-based methods.

This Research Design and Evaluation Protocol defines the research questions, evaluation metrics, paraphrase conditions, hard-negative examples, dataset splitting strategy and threshold-selection procedure. It also identifies the proposed contributions of the project.

The objective is to establish a systematic evaluation framework for comparing lexical and semantic plagiarism detection methods and developing an explainable system that provides relevant source passages and URLs as evidence.

Base Paper and Research Questions

Base Paper Details

Title: Overview of the Plagiarism Detection Task at PAN 2025

Authors: André Greiner-Petter, Maik Fröbe, Jan Philip Wahle, Terry Ruas, Bela Gipp, Akiko Aizawa and Martin Potthast.

Publication: CLEF 2025 Working Notes, CEUR Workshop Proceedings, Volume 4038.

arXiv ID: 2510.06805

Link: https://arxiv.org/html/2510.06805v1

Overview of the Base Paper

The PAN 2025 plagiarism detection task focuses on identifying plagiarism involving text rewritten using Large Language Models (LLMs). It provides a benchmark for evaluating different plagiarism detection approaches.

The benchmark contains 78,038 arXiv document pairs. The study demonstrates that embedding-based semantic approaches can achieve promising results in detecting paraphrased plagiarism. However, performance varies across datasets. The findings indicate that lexical and semantic methods have different strengths under different evaluation conditions.

This research motivates the investigation of both lexical and semantic similarity techniques in our project.

Research Questions

The following research questions define the main objectives of our project.

RQ1: How effectively can lexical and semantic methods detect exact copying, partial copying and different levels of paraphrasing?

Proposed target: Achieve an F1-score of at least 0.80.

RQ2: Does combining TF-IDF and SBERT improve plagiarism detection compared with individual methods?

Proposed target: Achieve at least a 3-percentage-point improvement in F1-score over the better individual method.

RQ3: Can the system distinguish actual plagiarism from hard-negative examples?

Proposed target: Achieve at least 90% precision on hard-negative examples.

RQ4: Can the system retrieve the original source and provide relevant passage evidence?

Proposed target: Retrieve the correct source within the top five results in at least 80% of applicable test cases.

Relevance to Our Project

The PAN 2025 study provides a foundation for investigating plagiarism detection under modern paraphrasing conditions. Based on its findings, our project will compare TF-IDF, MiniLM and MPNet using the PAN 2013 dataset for preliminary experiments.

The project will also investigate hard-negative detection and internet source checking to provide relevant matching passages and source URLs.

Expected Outcome

The expected outcome is a systematic comparison of lexical and semantic plagiarism detection approaches, identification of their strengths and limitations, and development of an explainable approach that provides evidence for detected plagiarism.

Evaluation Measures and Performance Metrics

Evaluation metrics are used to measure the effectiveness and accuracy of a plagiarism detection system. They help determine whether the system correctly identifies plagiarized passages and avoids incorrectly classifying original content as plagiarism.

In our project, Precision, Recall, F1-score, Accuracy, Granularity and PAN PlagDet will be considered for evaluating the performance of different plagiarism detection approaches.

Precision

Precision measures the proportion of passages predicted as plagiarized that are actually plagiarized.

$$Precision = \frac{TP}{TP + FP}$$

Where:

- TP (True Positive): Correctly identified plagiarized passages.
- FP (False Positive): Original passages incorrectly identified as plagiarized.

Higher precision indicates fewer false-positive predictions.

Recall

Recall measures the proportion of actual plagiarized passages correctly identified by the system.

$$Recall = \frac{TP}{TP + FN}$$

Where:

- TP (True Positive): Correctly identified plagiarized passages.
- FN (False Negative): Plagiarized passages missed by the system.

Higher recall indicates that fewer plagiarized passages are missed.

F1-Score

F1-score is the harmonic mean of Precision and Recall. It provides a balanced measure of both metrics.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

A higher F1-score indicates a better balance between detecting plagiarism and avoiding false positives.

Accuracy

Accuracy measures the proportion of all predictions that are correctly classified.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TN (True Negative): Original passages correctly identified as non-plagiarized.
- TP: Correctly identified plagiarized passages.
- FP: Incorrect plagiarism predictions.
- FN: Missed plagiarism cases.

Accuracy provides an overall measure of classification correctness.

Granularity

Granularity measures how many separate detections are required to cover one actual plagiarized passage.

A granularity value of 1 indicates that an actual plagiarism case is identified as one complete detection. Higher values indicate that the detected content is fragmented into multiple smaller passages.

Lower granularity is generally preferred because it represents more complete detections.

PAN PlagDet Score

PlagDet is an evaluation measure used in the PAN plagiarism detection framework. It considers both detection quality and granularity.

$$PlagDet = \frac{F1}{\log_2(1 + Granularity)}$$

Where:

- F1 represents the balance between Precision and Recall.
- Granularity measures fragmentation of detected passages.

A higher PlagDet score indicates better detection performance with less fragmentation.

Evaluation Strategy for Our Project

The following approaches will be evaluated:

1. TF-IDF: Measures lexical similarity based on shared words and terms.
2. MiniLM: Uses sentence embeddings to measure semantic similarity.

3. MPNet: Uses sentence embeddings to capture semantic relationships between passages.

The approaches will be compared using the same labelled evaluation data wherever applicable. Precision, Recall, F1-score and Accuracy will measure classification performance. Granularity and PAN PlagDet will be used for passage-level evaluation when the required reference annotations are available.

Reference

Potthast, M., Stein, B., Barrón-Cedeño, A., & Rosso, P. (2010). An Evaluation Framework for Plagiarism Detection. Proceedings of the 23rd International Conference on Computational Linguistics (COLING 2010).

Link: https://preview.aclanthology.org/fix_video/C10-2115.pdf

Paraphrase Conditions and Hard Negatives

Plagiarism can occur through direct copying as well as paraphrasing. In paraphrasing, the original text is rewritten using different words or sentence structures while retaining its meaning.

Our project considers different paraphrasing conditions and hard-negative examples to evaluate how effectively the system identifies plagiarism and distinguishes it from original content.

Paraphrase Conditions

The following four conditions will be considered during evaluation.

1. Exact Copying

Text copied directly from the original source without any changes.

Example:

- Original: Machine learning enables computers to learn patterns from data.
- Copied: Machine learning enables computers to learn patterns from data.

2. Partial Copying

Only a portion of the original passage is copied, while the remaining content may be different.

Example:

- Original: Machine learning enables computers to learn patterns from data and make predictions.
- Copied: Machine learning enables computers to learn patterns from data.

3. Light Paraphrasing

A few words are replaced with synonyms or minor changes are made to the sentence structure while retaining the original meaning.

Example:

- Original: Machine learning enables computers to learn patterns from data.
- Paraphrased: Machine learning allows computers to identify patterns from data.

4. Strong Paraphrasing

The vocabulary and sentence structure are significantly changed while retaining the original idea.

Example:

- Original: Machine learning enables computers to learn patterns from data.
- Paraphrased: Computer systems identify relationships in information to improve their predictions.

Hard Negatives

Hard negatives are independently written passages that may appear similar to the source because they contain related vocabulary or discuss the same topic, but are not derived from the source.

The following types will be considered:

1. Same-Topic Original Content: Independently written passages discussing the same subject.
2. Similar Vocabulary with Different Meaning: Passages sharing technical terms but communicating different information.
3. Independently Written Content: Original passages that may have high semantic similarity to the source.
4. PAWS Examples: Sentence pairs with high word overlap but different meanings, useful for testing false-positive predictions.

Evaluation Strategy

- Prepare labelled positive and negative passage pairs.
- Evaluate exact, partial, light and strong paraphrasing.
- Test TF-IDF, MiniLM and MPNet on the selected examples.
- Use hard negatives to identify false-positive predictions.

- Compare Precision, Recall and F1-score across different conditions.

The experiments are expected to demonstrate the strengths and limitations of lexical and semantic similarity methods.TF-IDF can identify shared words, while semantic models can identify meaning-related passages even when their wording differs. Hard-negative testing will help determine whether these methods incorrectly classify independently written content as plagiarism.

PAWS is a proposed source of additional hard-negative examples. Its inclusion in the final experimental dataset is yet to be confirmed.

Dataset Split Plan and Threshold Selection

Dataset splitting is an important step in evaluating a machine learning system. It divides the available data into separate subsets for development, threshold selection and final performance evaluation.

In our project, the PAN 2013 plagiarism detection dataset is used for preliminary experiments. The dataset is divided into training, validation and testing sets to support systematic evaluation of different similarity-based approaches.

Dataset Description

The PAN 2013 dataset provides plagiarism-related examples that can be used to evaluate plagiarism detection techniques.

For our preliminary experiments, the dataset is divided as follows:

Dataset	Number of Pairs	Purpose
Training Set	3,528	Development and experimentation
Validation Set	757	Threshold selection
Test Set	757	Final evaluation
Total	5,042	

Dataset Splitting

The dataset is divided into three subsets:

1. Training Set

The training set is used for initial experimentation and development of the similarity-based detection approaches.

2. Validation Set

The validation set is used to compare different similarity thresholds and select suitable values based on evaluation performance.

3. Test Set

The test set is reserved for final evaluation. It is used to measure the performance of the selected approaches on unseen data.

Similarity Threshold Selection

TF-IDF, MiniLM and MPNet calculate similarity scores between text passages. A similarity threshold is required to convert these scores into classification predictions.

The classification rule is:

$$Prediction = \begin{cases} 1, & Similarity \geq Threshold \\ 0, & Similarity < Threshold \end{cases}$$

Here, 1 represents a potential plagiarism match and 0 represents a non-match.

For example, if the selected threshold is 0.70:

- Similarity score 0.85: Potential plagiarism match.
- Similarity score 0.45: Non-match.

Threshold Selection Procedure

1. Calculate similarity scores for the passage pairs.
2. Experiment with different threshold values.
3. Evaluate the thresholds using validation data.
4. Compare Precision, Recall and F1-score.
5. Select the threshold that provides suitable validation performance.
6. Freeze the selected threshold before final testing.
7. Evaluate the selected approach using the test set.

Separate thresholds are selected for different similarity methods because their score distributions may differ.

Data Leakage Prevention

Data leakage can result in misleading evaluation performance. To minimize this risk:

- Maintain separate training, validation and test sets.
- Avoid overlap of source documents across different subsets wherever possible.

- Use validation data for threshold selection.
- Do not use test results to adjust thresholds.
- Keep the test set reserved for final evaluation.

Evaluation Strategy

The following approaches will be compared:

1. TF-IDF: Lexical similarity based on shared words and terms.
2. MiniLM: Semantic similarity using sentence embeddings.
3. MPNet: Semantic similarity using sentence embeddings.

Their performance will be evaluated using Precision, Recall, F1-score and Accuracy. Granularity and PAN PlagDet will also be considered where passage-level annotations are available.

The dataset splitting and threshold-selection strategy will help establish a consistent evaluation process. It will allow us to compare different similarity methods fairly and understand their performance on unseen examples.

Novelty Statement and Proposed Contributions

Novelty refers to the new features, improvements or contributions that distinguish our proposed project from existing research.

Our project, "Explainable Semantic Plagiarism Detection with Internet Source Checking," aims to improve plagiarism detection by evaluating lexical and semantic similarity methods and providing relevant source evidence for detected matches.

Existing Reference Work

6.2 Base Paper – PAN 2025

Paper: *Overview of the Plagiarism Detection Task at PAN 2025*

Authors: Greiner-Petter et al. (2025).

Publication: CLEF 2025 Working Notes, CEUR Workshop Proceedings, Volume 4038.

The PAN 2025 paper provides an overview of plagiarism detection research, including the challenges of detecting content paraphrased using large language models.

Key observations include:

- The evaluation involved a large collection of arXiv document pairs.
- Embedding-based cosine similarity achieved promising plagiarism detection performance.

- Performance varied across datasets, indicating that a method effective on one dataset may not perform equally well on another.
- The findings motivate further investigation of lexical and semantic approaches.

This paper provides the research background and motivation for our project.

Related Work – Semantic-Aware Plagiarism Detection

Paper: *Semantic-Aware Plagiarism Detection Using Machine Learning* (2025).

Authors: Gawade et al. (2025).

The existing system uses:

- TF-IDF for lexical feature extraction.
- SBERT for semantic representation.
- Cosine similarity for comparing sentences.
- Sentence-level plagiarism identification.
- Flask-based user interface.

This work provides a foundation for exploring lexical and semantic approaches to plagiarism detection.

Research Gap

Based on the PAN 2025 findings and the related semantic plagiarism detection approach, our project focuses on the following research areas:

1. Comparative evaluation of different lexical and semantic similarity models.
2. Testing detection performance across different paraphrasing conditions.
3. Evaluating false-positive predictions using hard-negative examples.
4. Identifying matching passages and providing relevant internet source URLs as evidence.
5. Improving the explainability of plagiarism detection through passage-level evidence.

Proposed Novelty

The proposed contributions of our project are:

1. Comparative Model Evaluation: Compare TF-IDF, MiniLM and MPNet to understand their effectiveness in detecting plagiarism.

2. Paraphrase-Based Evaluation: Evaluate exact copying, partial copying, light paraphrasing and strong paraphrasing.

3. Hard-Negative Testing: Test independently written, same-topic content to identify and evaluate false-positive predictions.

4. Internet Source Checking: Retrieve potential original sources from the internet and identify relevant matching passages.

5. Explainable Detection: Provide highlighted matching passages and source URLs to support plagiarism predictions.

Expected Contribution

The project aims to develop an explainable plagiarism detection system that combines comparative evaluation of lexical and semantic methods with internet source checking and evidence-based reporting

The PAN 2025 paper provides the research motivation, while the Semantic-Aware Plagiarism Detection paper provides a related existing approach for comparison. Our proposed work focuses on comparative model evaluation, paraphrase robustness, hard-negative testing and internet-based source evidence.

These contributions aim to improve the reliability and explainability of plagiarism detection.

References

[1] Greiner-Petter, A., Fröbe, M., Wahle, J. P., Ruas, T., Gipp, B., Aizawa, A., and Potthast, M. (2025). *Overview of the Plagiarism Detection Task at PAN 2025*. CLEF 2025 Working Notes, CEUR Workshop Proceedings, Volume 4038.

[2] Gawade, et al. (2025). *Semantic-Aware Plagiarism Detection Using Machine Learning*. ICSSSM 2025, Atlantis Press.